Graphnet Feed Spec – ADT HL7 - Generic

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Issue status	Version	Date	Actioned by	Description
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1 Purpose

The purpose of this document is to describe how CareCentric receives and displays ADT data through HL7 v2.4 messages.

1.1 Document Scope and Limitations

Any internal implementation details, XML Schemas and specifications defined by Graphnet within this document are subject to change without further notice. The CareCentric Highway module is beyond the scope of this document and any reference to the actual internal components or workings of this product in any form (e.g. visual, written) are subject to change without notice.

1.2 General Information / Reference Documents

The Graphnet interface specification is guided by the requirements of the following specifications and standards:
ANSI/HL7 V2.4-2000 (This is a revision of ANSI/HL7 V2.3.1-1999)

This document requires that the reader has a good understanding of HL7 messages, rules and format. Further information can be acquired from the following website:
https://lists.hl7.org.uk/standards/hl7-standards/hl7-version-2/

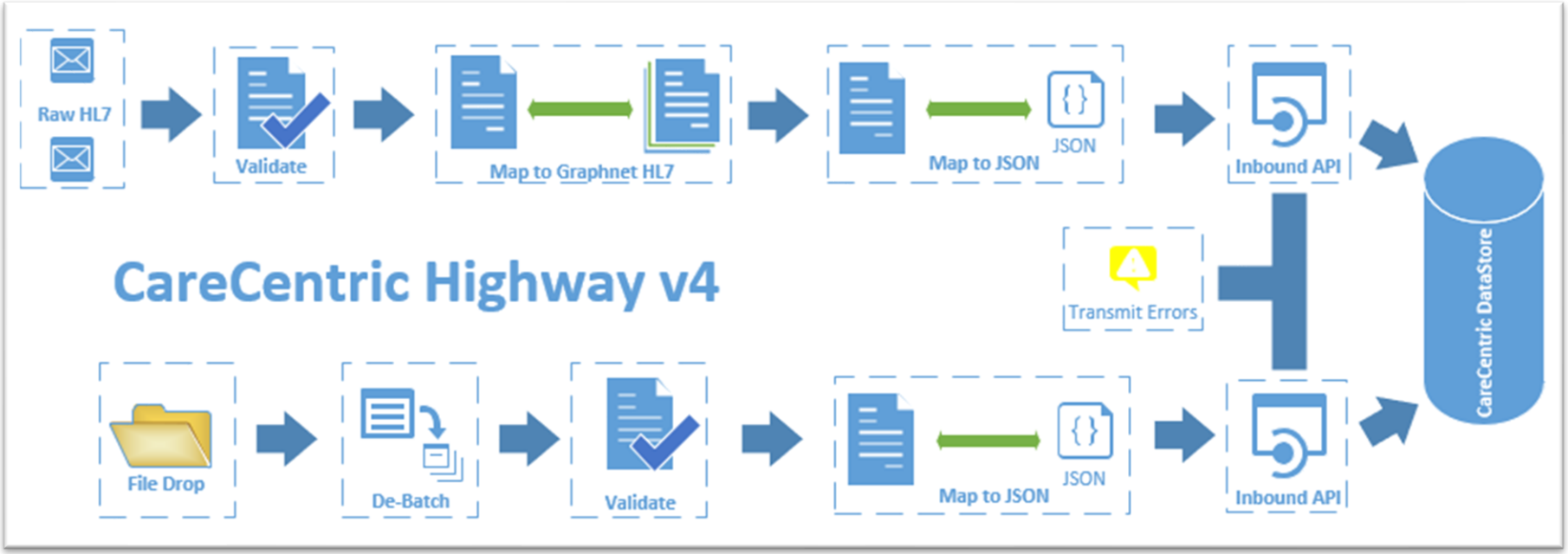
1.3 Definitions

Term/Abbreviation	Description
ACK	Acknowledgement
EHR	Electronic Health Record
HL7	Health Level Seven
MLLP	Minimum Lower Layer Protocol
Multi-Tenant System	CareCentric is a multi-tenancy system which is able to receive demographics information from more than one source and has the ability to link patient records based on specific patient matching criteria.
NAK	Negative Acknowledgement

2 CareCentric Highway Integration Engine

CareCentric Highway v4 is a cloud-based service where processing takes place in the Microsoft Azure platform. The service can also be deployed locally on premise. Data is sent to Highway using secure protocols HTTPS/FTPS. Azure meets a set of international and industry-specific compliance standards such as ISO/IEC 27001/27002:2013, HIPAA and FedRAMP.
The following diagram illustrates the concept for receiving data into CareCentric using the Highway Integration Engine to extract and integrate data from a wide range of health and social care IT solutions.

Note - This is not indicative of all physical environments and is provided as a conceptual guide only.



2.1 Transport/Network Security

Real Time HL7 messages are encrypted at socket level using TLS 1.2. Any messages other than HL7 will be encrypted when the files are sent to the Highway v4 server.
The configuration of the network including firewalls, IP addresses and re-routing of data are the responsibility of the sending party (Trust).

2.2 Error Handling

Highway v4 incorporates synchronous ACK/NACK Reponses when HL7 messages are received. If any issues occur during processing, they are logged (as per diagram above) and can be configured to be sent out as Email Notifications.

3 Message Format

3.1 HL7 Supported Message Types for ADT

The following HL7 messages are accepted by CareCentric Highway to display ADT information for a patient:

HL7 Message Type	Event/Description
ADT^A01	Admit a patient
ADT^A02	Transfer a patient
ADT^A03	Discharge a patient
ADT^A04	Register a patient
ADT^A05	Pre-admit a patient
ADT^A08	Update patient information
ADT^A11	Cancel admit
ADT^A12	Cancel transfer
ADT^A13	Cancel discharge
ADT^A28	Add person information
ADT^A31	Update person information
ADT^A38	Cancel pre-admit
ADT^A40	Merge patient information – PID

4 HL7 Message Details

4.1 Mandatory Fields

4.1.1 Message Type [MSH-9]

Each HL7 message received will contain a value in the *Message Type* field which is passed into the following parameter: [MSH-9].
This is used to help identify the type of message being received (see section 3.1).

4.1.2 Event Reason Code [EVN-4]

The event reason code is used to further differentiate between HL7 messages where the Message Type is used to represent similar although different messages. Currently, only Inpatient messages have this requirement.

Message	Message Type	Event Reason Code	Meaning	Patient Class	Unique Code
Inpatient Admission	A01	100	Inpatient Admission	Inpatient	
	A08	101	Update Inpatient Admission		
Inpatient Discharge	A03	140	Inpatient Discharge		
	A08	141	Update Inpatient Discharge		
Inpatient Transfer	A02	120	Ward Transfer		
		130	Consultant/Specialty Transfer		
	A08	121	Update Ward Transfer		Contains "-" but doesn't start with or end with "-"
		131	Consultant/Specialty Transfer		
Inpatient Admission Cancellation	A11	102	Cancel Inpatient Admission		
Patient Discharge Cancellation	A13	142	Cancel Discharge		
Inpatient Transfer Cancellation	A12	122	Cancel Ward Transfer		
		132	Cancel Consultant/Specialty Transfer		

4.1.3 Patient Visit Class [PV1-2]

Each HL7 message received will contain a value in the *Patient Visit Code* Mapping field which is passed into the following parameter: [**PV1-2]**.
This is used to help identify whether the message relates to an Inpatient, Outpatient or Accident & Emergency can be determined based on the following values:

Message Type	Patient Visit Code Mapping [PV1-2]
Accident & Emergency	**AE** or **E**
Inpatient	**I**
Outpatient	**O**
Waiting List	**W**

4.1.4 Message Control ID [MSH-10]

Each HL7 message received should contain a value in the *Message Control ID* field which is passed into the following parameter: [**MSH-10**].
This value will be used as the unique ID for the message and any messages with the same value will be treated as a new version. This unique ID is used for messaging processes only (Transport layer verification).

4.1.5 Visit Number [PV1-19] / Pre-Admit Number [PV1-5]

Each HL7 message received should contain a value in the *Visit Number* field which is passed into the following parameter: [**PV1-19**].
If [**PV1-5**] is present, this is instead used as the key identifier for version control and updating documents in CareCentric.

Note: For referral (REF) messages, the External Referral Identifier field [**RF1-11**] can be used as a suitable alternative for version control.

4.1.6 Message Sending Facility [MSH-4]

Each HL7 message received should contain a value in the *Message Sending Facility* field which is passed into the following parameter:[**MSH-4**].
This value will be used to display the name of the source system in CareCentric against the patients ADT and Referral information.

4.1.7 Message Date Time [MSH-7]

Each HL7 message received should contain a value in the *Date/Time of Message* field which is passed into the following parameter:[**MSH-7**].
This value will be used to determine the date of the event for the message type to display in CareCentric against the patient's record.

The value of a point in time is represented using the ISO 8601 compliant form traditionally in use with HL7. This is the form that has no decorating dashes, colons and no "T" between the date and time. The syntax is "YYYYMMDDHHMMSS.UUUU[+|-ZZzz]" where digits can be omitted from the right side to express less precision. Common forms are "YYYYMMDD" and "YYYYMMDDHHMM", but the ability to truncate on the right side is not limited to these two variants. E.g. 20150210110056393+0000.

4.1.8 Patient Identifier List [PID-3/PID-3.5]

Each HL7 message received should contain a value in the *Patient Identifier List* field which is passed into the following parameter:[**PID-3**].
The value passed in here should be the patient's local number and / or NHS number. The number type should be indicated by the *Identifier Type Code* in PID 3.5.

CareCentric stores the NHS number and an additional number such as a PAS number or other Patient ID from the trust.
Supported Number Types and Codes to use:

Identifier Type	Numbers
NHS / NH	NHS number
PAS / PI	Local System number or PAS Number or Hospital Number
MRN	Master Record Number
HOSP	Hospital Number
iPM ID	

4.1.9 Patient Name selection [PID-5.7] = L

The patient name is selected by checking for the Legal name tag "**L**" in sub-field 7 of field 5. If no Legal tag exists, then the first available name will be chosen.

4.1.10 Home address selection [PID-11.7] = H

The patient home address is selected by checking for the Home tag "**H**" in sub-field 7 of field 11. If no home tag exists, then the first available address will be chosen.

4.1.11 Demographics contact details

Demographics details can be obtained by searching for the following values.

Demographic detail	Test Field	Field Value
Home Telephone	PID-13.3 = “HOME” and PID-13.2 = “PH” [if 13.2 is populated]	PID-13.1
Mobile Telephone	PID-13.3 = “MC” and PID-13.2 = “PH” [if 13.2 is populated]	PID-13.1
Work Telephone	PID-14.3 = “WP” and PID-14.2 = “PH” [if 14.2 is populated]	PID-14.1
Email Address	PID-13.2 = “Internet”	PID-14.1

4.1.12 HL7 Message Content Validation

If data being sent via HL7 contains any null references or empty strings, then the message can either contain NULL or no data at all.
E.g. If deceased date has no value, then this should be sent across as an empty field. The following would be valid:

- |NULL|
- ||

The HL7 message should escape any reserved characters. These typically are:

- ^
- ~
- \
- &
- |

If any data contains these characters, they will need to be escaped as described in the HL7 v2.4 specifications set out by the HL7 standards.

4.2 Patient Matching Rules

Patients will be matched based on the Patient Number as defined in PID.3.1 (PID.3.1 is a field within the message). The Tenancy ID is used to accurately match information related to a specific tenancy, distinct from any other tenancy. Full details on Patient Matching & Tenancy Linking can be found in the Graphnet Customer Library.

4.3 HL7 Message Data and Codes

CareCentric by default display the data extracted from the HL7 message in the format it is sent. Therefore, any locally used codes for descriptions need to be translated before being passed into the message and out to CareCentric.
CareCentric does have the ability if required to perform look-ups on codes when they are received in a HL7 message. This is the role of the Reference Data Service (**RDS**). If RDS is to be used for code lookups, this will require a separate review of the requirements.

4.4 Message Details - Key

[ ] - Brackets used will indicate that the segment is optional.
{ } - Braces used will indicate that the segment can be repeated.

4.5 Message Details – Accident & Emergency

4.5.1 A&E – Attendance (ADT^A04)

A&E attendance messages are accepted by Graphnet as Message Type ADT^A04.
ADT^A04 Message Structure:

Segments	Description	Comments
MSH	Message Header	
EVN	Event Type	
PID	Patient Identifiable Data	Only used for patient matching
[PD1]	Additional Demographic	
PV1	Patient Visit	
[PV2]	Additional Patient Visit	
[{ AL1 }]	Allergy Information	
[{ DG1 }]	Diagnostic Information	If a PR1 segment is being sent, and there is no Diagnosis information, we still require an empty DG1 segment.
[{ PR1 }]	Procedures	If a DG1 segment is being sent, and there is no Procedure information, we still require an empty PR1 segment.

The following fields are used by this transaction type for patient and activity matching purposes:

Segment	Field	Description	Comments
EVN	2.1	EventRecordedDate/Time	
EVN	6.1	EventOccurredDate/Time	
PID	3.1	PatientIDList/PatientID	
PID	3.5	Patient Identifier Type	Used to determine whether the number is NHS type or other (see section 4.1.8)
PV1	2.1	PatientClass	Must be set as 'AE' or 'E'
PV1	19.1	EventIdentifier	

4.5.2 A&E – Update Patient Information (ADT^A08)

A&E update messages are accepted by Graphnet as Message Type ADT^A08. An A08 update message is treated as an update to an A&E Admission (A01) where visit class is set to 'AE' or 'E'.

ADT^A08 Message Structure:

Segments	Description	Comments
MSH	Message Header	
EVN	Event Type	
PID	Patient Identifiable Data	Only used for patient matching
[PD1]	Additional Demographic	
PV1	Patient Visit	
[PV2]	Additional Patient Visit	
[{ AL1 }]	Allergy Information	
[{ DG1 }]	Diagnosis Information	If a PR1 segment is being sent, and there is no Diagnosis information, we still require an empty DG1 segment.
[{ PR1 }]	Procedures	If a DG1 segment is being sent, and there is no Procedure information, we still require an empty PR1 segment.

The following fields are used by this transaction type for patient and activity matching purposes:

Segment	Field	Description	Comments
EVN	2.1	EventRecordedDate/Time	
EVN	6.1	EventOccurredDate/Time	
PID	3.1	PatientIDList/PatientID	
PID	3.5	Patient Identifier Type	Used to determine whether the number is NHS type or other (see section 4.1.8)
PV1	2.1	PatientClass	Must be set as 'AE' or 'E'
PV1	19.1	EventIdentifier	

4.5.3 A&E – Discharge (ADT^A03)

A&E discharge messages are accepted by Graphnet as Message Type ADT^A03.

ADT^A03 Message Structure:

Segments	Description	Comments
MSH	Message Header	
EVN	Event Type	
PID	Patient Identifiable Data	Only used for patient matching
[PD1]	Additional Demographic	
PV1	Patient Visit	
[DG1]	Diagnosis Information	If a PR1 segment is being sent, and there is no Diagnosis information, we still require an empty DG1 segment.
[{ PR1 }]	Procedures	If a DG1 segment is being sent, and there is no Procedure information, we still require an empty PR1 segment.

The following fields are used by this transaction type for patient and activity matching purposes:

Segment	Field	Description	Comments
EVN	2.1	EventRecordedDate/Time	
EVN	6.1	EventOccurredDate/Time	
PID	3.1	PatientIDList/PatientID	
PID	3.5	Patient Identifier Type	Used to determine whether the number is NHS type or other (see section 4.1.8)
PV1	2.1	PatientClass	Must be set as 'AE' or 'E'
PV1	19.1	EventIdentifier	

4.5.4 A&E – Cancel Attendance (ADT^A11)

A&E attendance cancellation messages are accepted by Graphnet as Message Type ADT^A11. Upon receipt of an A11 the corresponding A&E attendance document will be removed from the Patient Record.

ADT^A11 Message Structure:

Segments	Description	Comments
MSH	Message Header	
EVN	Event Type	
PID	Patient Identifiable Data	Only used for patient matching
[PD1]	Additional Demographics	
PV1	Patient Visit	

The following fields are used by this transaction type for patient and activity matching purposes:

Segment	Field	Description	Comments
EVN	2.1	EventRecordedDate/Time	
EVN	6.1	EventOccurredDate/Time	
PID	3.1	PatientIDList/PatientID	
PID	3.5	Patient Identifier Type	Used to determine whether the number is NHS type or other (see section 4.1.8)
PV1	2.1	PatientClass	Must be set as 'AE' or 'E'
PV1	19.1	EventIdentifier	To identify the activity document to remove

4.5.5 A&E – Cancel Discharge (ADT^A13)

A&E discharge cancellation messages are accepted by Graphnet as Message Type ADT^A13. Upon receipt of an A13 the corresponding A&E Discharge document will be removed from the Patient Record.

ADT^A13 Message Structure:

Segments	Description	Comments
MSH	Message Header	
EVN	Event Type	
PID	Patient Identifiable Data	Only used for patient matching
[PD1]	Additional Demographics	
PV1	Patient Visit	
[PV2]	Additional Patient Visit	

The following fields are used by this transaction type for patient and activity matching purposes:

Segment	Field	Description	Comments
EVN	2.1	EventRecordedDate/Time	
EVN	6.1	EventOccurredDate/Time	

4.6 Inpatient – Admission (ADT^A01)

IP admission messages are accepted by Graphnet as Message Type ADT^A01.

ADT^A01 Message Structure:

Segments	Description	Comments
MSH	Message Header	
EVN	Event Type	
PID	Patient Identifiable Data	Only used for patient matching
[PD1]	Patient Additional Demographic	
PV1	Patient Visit	
[PV2]	Additional Patient Visit	
[{ AL1 }]	Allergy Information	
[{ DG1 }]	Diagnosis Information	If a PR1 segment is being sent, and there is no Diagnosis information, we still require an empty DG1 segment.
[{ PR1 }]	Procedures	If a DG1 segment is being sent, and there is no Procedure information, we still require an empty PR1 segment.

The following fields are used by this transaction type for patient and activity matching purposes:

Segment	Field	Description	Comments
EVN	2.1	EventRecordedDate/Time	
EVN	6.1	EventOccurredDate/Time	
PID	3.1	PatientIDList/PatientID	
PID	3.5	Patient Identifier Type	Used to determine whether the number is NHS type or other (see section 4.1.8)
PV1	2.1	PatientClass	Must be set as 'I'
PV1	19.1	EventIdentifier	

4.6.1 Inpatient – Transfer (ADT^A02)

IP transfer messages are accepted by Graphnet as Message Type ADT^A02.

ADT^A02 Message Structure:

Segments	Description	Comments
MSH	Message Header	
EVN	Event Type	
PID	Patient Identifiable Data	Only used for patient matching
[PD1]	Additional Demographic	
PV1	Patient Visit	
[PV2]	Additional Patient Visit	

The following fields are used by this transaction type for patient and activity matching purposes:

Segment	Field	Description	Comments
EVN	2.1	EventRecordedDate/Time	
EVN	6.1	EventOccurredDate/Time	
PID	3.1	PatientIDList/PatientID	
PID	3.5	Patient Identifier Type	Used to determine whether the number is NHS type or other (see section 4.1.8)
PV1	2.1	PatientClass	Must be set as 'I'
PV1	19.1	EventIdentifier	

4.6.2 Inpatient – Discharge (ADT^A03)

IP discharge messages are accepted by Graphnet as Message Type ADT^A03.

ADT^A03 Message Structure:

Segments	Description	Comments
MSH	Message Header	
EVN	Event Type	
PID	Patient Identifiable Data	Only used for patient matching
[PD1]	Additional Demographics	
PV1	Patient Visit	
[PV2]	Additional Patient Visit	
[{ AL1 }]	Allergy Information	
[{ DG1 }]	Diagnosis Information	If a PR1 segment is being sent, and there is no Diagnosis information, we still require an empty DG1 segment.
[{ PR1 }]	Procedures	If a DG1 segment is being sent, and there is no Procedure information, we still require an empty PR1 segment.

The following fields are used by this transaction type for patient and activity matching purposes:

Segment	Field	Description	Comments
EVN	2.1	EventRecordedDate/Time	
EVN	6.1	EventOccurredDate/Time	
PID	3.1	PatientIDList/PatientID	
PID	3.5	Patient Identifier Type	Used to determine whether the number is NHS type or other (see section 4.1.8)
PV1	2.1	PatientClass	Must be set as '**I**'
PV1	19.1	EventIdentifier	

4.6.3 Inpatient – Update Patient Information (ADT^A08)

IP update messages are accepted by Graphnet as Message Type ADT^A08. An A08 update message is treated as an update to an Inpatient Admission (A01) where visit class is set to 'I'.

ADT^A08 Message Structure:

Segments	Description	Comments
MSH	Message Header	
EVN	Event Type	
PID	Patient Identifiable Data	Only used for patient matching
[PD1]	Additional Demographic	
PV1	Patient Visit	
[PV2]	Additional Patient Visit	
[{ AL1 }]	Allergy Information	
[{ DG1 }]	Diagnosis Information	If a PR1 segment is being sent, and there is no Diagnosis information, we still require an empty DG1 segment.
[{ PR1 }]	Procedures	If a DG1 segment is being sent, and there is no Procedure information, we still require an empty PR1 segment.

The following fields are used by this transaction type for patient and activity matching purposes:

Segment	Field	Description	Comments
EVN	2.1	EventRecordedDate/Time	
EVN	6.1	EventOccurredDate/Time	
PID	3.1	PatientIDList/PatientID	
PID	3.5	Patient Identifier Type	Used to determine whether the number is NHS type or other (see section 4.1.8)
PV1	2.1	PatientClass	Must be set as 'I'
PV1	19.1	EventIdentifier	

4.6.4 Inpatient – Cancel Admission (ADT^A11)

IP admission cancellation messages are accepted by Graphnet as Message Type ADT^A11. Upon receipt of an A11 the corresponding IP attendance document will be removed from the Graphnet EPR.

ADT^A11 Message Structure:

Segments	Description	Comments
MSH	Message Header	
EVN	Event Type	
PID	Patient Identifiable Data	Only used for patient matching
[PD1]	Additional Demographics	
PV1	Patient Visit	
[PV2]	Additional Patient Visit	

The following fields are used by this transaction type for patient and activity matching purposes:

Segment	Field	Description	Comments
EVN	2.1	EventRecordedDate/Time	
EVN	6.1	EventOccurredDate/Time	
PID	3.1	PatientIDList/PatientID	
PID	3.5	Patient Identifier Type	Used to determine whether the number is NHS type or other (see section 4.1.8)
PV1	2.1	PatientClass	Must be set as 'I'
PV1	19.1	EventIdentifier	

4.6.5 Inpatient – Cancel Transfer (ADT^A12)

IP transfer cancellation messages are accepted by Graphnet as Message Type ADT^A12. Upon receipt of an A12 the corresponding IP attendance document will be removed from the Patient Record.

ADT^A12 Message Structure:

Segments	Description	Comments
MSH	Message Header	
EVN	Event Type	
PID	Patient Identifiable Data	Only used for patient matching
[PD1]	Additional Demographics	
PV1	Patient Visit	
[PV2]	Additional Patient Visit	

The following fields are used by this transaction type for patient and activity matching purposes:

Segment	Field	Description	Comments
EVN	2.1	EventRecordedDate/Time	
EVN	6.1	EventOccurredDate/Time	
PID	3.1	PatientIDList/PatientID	
PID	3.5	Patient Identifier Type	Used to determine whether the number is NHS type or other (see section 4.1.8)
PV1	2.1	PatientClass	Must be set as 'I'
PV1	19.1	EventIdentifier	

4.6.6 Inpatient – Cancel Discharge (ADT^A13)

IP discharge cancellation messages are accepted by Graphnet as Message Type ADT^A13. Upon receipt of an A13 the corresponding IP attendance document will be removed from the Patient Record.

ADT^A13 Message Structure:

Segments	Description	Comments
MSH	Message Header	
EVN	Event Type	
PID	Patient Identifiable Data	Only used for patient matching
[PD1]	Additional Demographics	
PV1	Patient Visit	
[PV2]	Additional Patient Visit	

The following fields are used by this transaction type for patient and activity matching purposes:

Segment	Field	Description	Comments
EVN	2.1	EventRecordedDate/Time	
EVN	6.1	EventOccurredDate/Time	
PID	3.1	PatientIDList/PatientID	
PID	3.5	Patient Identifier Type	Used to determine whether the number is NHS type or other (see section 4.1.8)
PV1	2.1	PatientClass	Must be set as 'I'
PV1	19.1	EventIdentifier	

4.7 Outpatient – Referral (REF^I12/I13)

OP referral\referral update messages are accepted by Graphnet as Message Type REF^I12\I13 messages respectively.

REF^I12/I13 Message Structure:

Segments	Description	Comments
MSH	Message Header	
RF1	Referral Information	
[{ PRD }]	Provider Data	
PID	Patient Identification	Used for patient matching only
[PD1]	Additional Demographic	
[{ PV1 }]	Patient Visit Information	
[{ PV2 }]	Additional Patient Visit	
[{ NTE }]	Notes and Comments	

The following fields are used by this transaction type for patient and activity matching purposes:

Segment	Field	Description	Comments
RF1	9.1	ProcessDate	
RF1	7.1	EffectiveDate	
PID	3.1	PatientIDList/PatientID	
PID	3.5	Patient Identifier Type	Used to determine whether the number is NHS type or other (see section 4.1.8)
RF1	6.1	Referral Identifier	

4.7.1 Outpatient – Close/Cancel Referral (REF^I14)

OP referral closure messages are accepted by Graphnet as Message Type REF^I4. Upon receipt of a REF^I14 the corresponding OP referral document will be removed from the Patient Record.

REF^I14Message Structure:

Segments	Description	Comments
MSH	Message Header	
RF1	Referral Information	
[{ PRD }]	Provider Data	
PID	Patient Identification	Used for patient matching only
[PD1]	Additional Demographic	

The following fields are used by this transaction type for patient and activity matching purposes:

Segment	Field	Description	Comments
RF1	10.1	ProcessDate	
RF1	9.1	EffectiveDate	
PID	3.1	PatientIDList/PatientID	
PID	3.5	Patient Identifier Type	Used to determine whether the number is NHS type or other (see section 4.1.8)
RF1	2.1	ReferralType	Must be set as 'O'
RF1	6.1	Referral Identifier	

4.7.2 Outpatient – Attendance (ADT^A01/A04)

Outpatient attendance messages are accepted by Graphnet as Message Type ADT^A01 (or A04 with visit class set as '**O**').

ADT^A01 or A04 Message Structure:

Segments	Description	Comments
MSH	Message Header	
EVN	Event Type	
PID	Patient Identifiable Data	Only used for patient matching
[PD1]	Additional Demographics	
PV1	Patient Visit	
[PV2]	Additional Patient Visit	
[{ AL1 }]	Allergy Information	
[{ DG1 }]	Diagnosis Information	If a PR1 segment is being sent, and there is no Diagnosis information, we still require an empty DG1 segment.
[{ PR1 }]	Procedures	If a DG1 segment is being sent, and there is no Procedure information, we still require an empty PR1 segment.

The following fields are used by this transaction type for patient and activity matching purposes:

Segment	Field	Description	Comments
EVN	2.1	EventRecordedDate/Time	
EVN	6.1	EventOccurredDate/Time	
PID	3.1	PatientIDList/PatientID	
PID	3.5	Patient Identifier Type	Used to determine whether the number is NHS type or other (see section 4.1.8)
PV1	2.1	PatientClass	Must be set as '**O**'
PV1	19.1	EventIdentifier	

4.7.3 Outpatient – Update Patient Information (ADT^A08)

Outpatient update messages are accepted by Graphnet as Message Type ADT^A08. An A08 update message is treated as an update to an Outpatient Admission (A01) where visit class is set to '**O**'.

ADT^A08 Message Structure:

Segments	Description	Comments
MSH	Message Header	
EVN	Event Type	
PID	Patient Identifiable Data	Only used for patient matching
[PD1]	Additional Demographics	
PV1	Patient Visit	
[PV2]	Additional Patient Visit	
[{ AL1 }]	Allergy Information	
[{ DG1 }]	Diagnosis Information	If a PR1 segment is being sent, and there is no Diagnosis information, we still require an empty DG1 segment.
[{ PR1 }]	Procedures	If a DG1 segment is being sent, and there is no Procedure information, we still require an empty PR1 segment.

The following fields are used by this transaction type for patient and activity matching purposes:

Segment	Field	Description	Comments
EVN	2.1	EventRecordedDate/Time	
EVN	6.1	EventOccurredDate/Time	
PID	3.1	PatientIDList/PatientID	
PID	3.5	Patient Identifier Type	Used to determine whether the number is NHS type or other (see section 4.1.8)
PV1	2.1	PatientClass	Must be set as '**O**'
PV1	19.1	EventIdentifier	

4.7.4 Outpatient – Discharge (ADT^A03)

Outpatient attendance messages are accepted by Graphnet as Message Type ADT^A03.

ADT^A03 Message Structure:

Segments	Description	Comments
MSH	Message Header	
EVN	Event Type	
PID	Patient Identifiable Data	Only used for patient matching
[PD1]	Additional Demographics	
PV1	Patient Visit	
[PV2]	Additional Patient Visit	
[{ AL1 }]	Allergy Information	
[{ DG1 }]	Diagnosis Information	If a PR1 segment is being sent, and there is no Diagnosis information, we still require an empty DG1 segment.
[{ PR1 }]	Procedures	If a DG1 segment is being sent, and there is no Procedure information, we still require an empty PR1 segment.

The following fields are used by this transaction type for patient and activity matching purposes:

Segment	Field	Description	Comments
EVN	2.1	EventRecordedDate/Time	
EVN	6.1	EventOccurredDate/Time	
PID	3.1	PatientIDList/PatientID	
PID	3.5	Patient Identifier Type	Used to determine whether the number is NHS type or other (see section 4.1.8)
PV1	2.1	PatientClass	Must be set as '**O**'
PV1	19.1	EventIdentifier	

4.7.5 Future Admissions (ADT^A05)

Pre-Attendance messages are accepted by Graphnet as Message Type ADT^A05.

ADT^A05 Message Structure:

Segments	Description	Comments
MSH	Message Header	
EVN	Event Type	
PID	Patient Identifiable Data	Only used for patient matching
[PD1]	Additional Demographics	
PV1	Patient Visit	
[PV2]	Additional Patient Visit	
[{ AL1 }]	Allergy Information	
[{ DG1 }]	Diagnosis Information	If a PR1 segment is being sent, and there is no Diagnosis information, we still require an empty DG1 segment.
[{ PR1 }]	Procedures	If a DG1 segment is being sent, and there is no Procedure information, we still require an empty PR1 segment.

The following fields are used by this transaction type for patient and activity matching purposes:

Segment	Field	Description	Comments
EVN	2.1	EventRecordedDate/Time	
EVN	6.1	EventOccurredDate/Time	
PID	3.1	PatientIDList/PatientID	
PID	3.5	Patient Identifier Type	Used to determine whether the number is NHS type or other (see section 4.1.8)
PV1	2.1	PatientClass	Must be set as '**O**' for Outpatient and '**W**' for Waiting List activity types
PV1	5	Preadmit Number	

4.7.6 Cancel Future Admission (ADT^A38)

Pre-Attendance cancellation messages are accepted by Graphnet as Message Type ADT^A38.

ADT^A38 Message Structure:

Segments	Description	Comments
MSH	Message Header	
EVN	Event Type	
PID	Patient Identifiable Data	Only used for patient matching
[PD1]	Additional Demographics	
PV1	Patient Visit	
[PV2]	Additional Patient Visit	

The following fields are used by this transaction type for patient and activity matching purposes:

Segment	Field	Description	Comments
EVN	2.1	EventRecordedDate/Time	
EVN	6.1	EventOccurredDate/Time	
PID	3.1	PatientIDList/PatientID	
PID	3.5	Patient Identifier Type	Used to determine whether the number is NHS type or other (see section 4.1.8)
PV1	2.1	PatientClass	Must be set as '**O**' for Outpatient and '**W**' for Waiting List activity types
PV1	5/19.1	Preadmit number/EventIdentifier	

4.8 Patient Demographics

Patient demographic information can be created / updated from the following message types:

- A31
- A01
- A04
- A05
- A08
- REF
- A40

4.8.1 Demographics – New Registration (ADT^A28)

This message type will create a patient if they do not exist in the Master Patient Index. If a match is found (using data from the PID segment), then no updates or further action will be taken. If a match is not found, the message must include all the required fields.

New Registration messages are accepted by Graphnet as Message Type ADT^A28.

ADT^A28 Message Structure:

Segments	Description	Comments
MSH	Message Header	
EVN	Event Type	
PID	Patient Identifiable Data	
[PD1]	Additional Demographics	

The following fields are used by this transaction type for patient and activity matching purposes:

Segment	Field	Description	Comments
PID	3.1	PatientIDList/PatientID	
PID	3.5	Patient Identifier Type	Used to determine whether the number is NHS type or other (see section 4.1.8)

4.8.2 Demographics – Update (ADT^A31)

This message type will update patient information if they exist in the Master Patient Index. If a match is found, an update to their demographic information will be made. Previous patient identifier information will be deleted with the information from the update. If no patient is found, the message will go through the retry process and if no match is found the message will either fail or create the patient depending based on the Patient Matching/Tenancy Linking rules in CareCentric.

Patient Update messages are accepted by Graphnet as Message Type ADT^A31.

ADT^A31 Message Structure:

Segments	Description	Comments
MSH	Message Header	
EVN	Event Type	
PID	Patient Identifiable Data	
[PD1]	Additional Demographics	
[PV1]	Patient Visit	

The following fields are used by this transaction type for patient and activity matching purposes:

Segment	Field	Description	Comments
PID	3.1	PatientIDList/PatientID	
PID	3.5	Patient Identifier Type	Used to determine whether the number is NHS type or other (see section 4.1.8)

4.8.3 Demographics – Merge (ADT^A40) / (ADT^A34)

This message type will merge patients which exist in the Master Patient Index. If a patient match is found in both the MRG and PID segment, an update to their demographic information will be made where the previous information from the MRG segment will be merged with the new information available in the PID segment. The Master Patient Index will record the patient as a merged patient record. If a match cannot be made on the patient identifiable data in the MRG and / or PID segment, the message will go through the retry process and if no match is found the message will fail.

Patient Update messages are accepted by Graphnet as Message Type ADT^A40 or ADT^A34.

ADT^A40 / ADT^A34 Message Structure:

Segments	Description	Comments
MSH	Message Header	
EVN	Event Type	
PID	Patient Identifiable Data	
[PD1]	Additional Demographics	
MRG	Merge	

The following fields are used by this transaction type for patient and activity matching purposes:

Segment	Field	Description	Comments
PID	3.1	PatientIDList/PatientID	
PID	3.5	Patient Identifier Type	Used to determine whether the number is NHS type or other (see section 4.1.8)
MRG	1.1	PatientIDList/PatientID	
MRG	1.1	Patient Identifier Type	Used to determine whether the number is NHS type or other (see section 4.1.8)

4.9 Allergies & Alerts

Graphnet can process Allergies and/or Alerts where the inbound message contains the AL1 segment and it is supported by the appropriate transaction within CareCentric Highway. If either Allergies and/or Alerts are to be supported, then Allergies must use an identifying prefix/code to differentiate.

Note: The prefix/code must be confirmed with Graphnet.

When being provided with Allergy and Alert information, each message will need to contain all the latest available number of Allergies/Alerts for a patient (each applicable HL7 message will require an AL1 segment). CareCentric will clear down and update any Allergy and/or Alert information with the latest inbound message. Therefore as an example, if the latest inbound message does not contain any allergies or alerts, any existing data for that patient will be cleared down. This also applies to partial updates where all allergies and alerts will be cleared down and the partial updates taken as the latest allergy and alert information for a patient.

5 Segment Details

5.1 MSH – Message Header

Seq	Field Name	Opt	Occurs	Comments
1	Field Separator	R	1	
2	Encoding Characters	R	1	
3	Sending Application	R	1	Used to generate document properties
4	Sending Facility	O	1	Used to generate document properties
5	Receiving Application	R	1	
6	Receiving Facility	O	1	
7	Date/Time of Message	R	1	
9.1	MessageType\Message Type	R	1	
9.2	MessageType\TriggerEvent	R	1	Used to generate document properties
10.1	MessageControlID	R	1	Used to generate document properties
11	ProcessingID	O	1	
12	VersionID	O	1	
18	Character Set	O	1	

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5.2 EVN – Event Information

Seq	Field Name	Opt/Req	Occurs	Comments
1	Event Type Code	O	1	
2	Recorded Date/Time	R	1	
3	Date/Time Planned	O	1	
4	Event Reason Code	O	1	
5	Operator ID	O	1	
6	Event Occurred	R	1	
7	Event Facility	O	1	

5.3 PID – Patient Identifiable Data Segment

Seq	Field Name	Opt/Req	Occurs	Comments
1	Set ID	O	1	Used to retrieve NHS number and/or Other number type (to be used with identifier type)
2	Patient ID	R*	1	
3.1	PatientIdentifierList\PatientID	R	1-n	Used to retrieve NHS number and/or Other number type (to be used with identifier type)
3.4	PatientIdentifierList\AssigningAuthority	O	1-n	
3.5	Identifier Type Code	R	1-n	Used to identify number type **NHS** / **NH** – NHS Number **PAS** / **PI** - PAS or Hospital Number **HOSP** – Hospital Number **MRN** – Master Record Number
5.1	Surname	R	1-n	If value can't be provided, send the value 'Not Known'
5.2	Forename	R	1-n	If value can't be provided, send the value 'Not Known'
5.3	Middle Name(s)	O	1-n	

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Seq	Field Name	Opt/Req	Occurs	Comments
5.4	Suffix	O	1-n	
5.5	Prefix	R	1-n	e.g. **Master**, **Mr**, **Mrs**
5.7	Name Type Code	O	1-n	**L** = Legal Name Anything other than **L** is ignored. If no value is supplied in any of the repeats then the first repeat is used.
7	Date of Birth	R	1	e.g. **YYYYMMDDHHMM**
8	Administrative Sex	R	1	Code values accepted: **F**- Female **M**-Male **O** or **U** = Unknown
9	Patient Alias	O	1-n	
11	Patient Address	R	1-n	If value can't be provided, send the value 'Not Known'
11.1	Address	R		If value can't be provided, send the value 'Not Known'
11.2	Street	R		If value can't be provided, send the value 'Not Known'
11.3	Town/City	O		
11.4	County	O		
11.5	Post Code	O		
11.6	Country	O		
11.7	Address Type	O		**H** = Home Anything other than **H** is ignored. If no value is supplied in any of the repeats
11.12	Address Date	O		
12	County/Parish Code	O	1	
13	Home Phone Number	O	1-n	See section on Demographics data
13.1	Number	O	1	
13.2	Telecoms Use code	O		e.g. **PH**, **Internet**

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Seq	Field Name	Opt/Req	Occurs	Comments
13.3	Telecoms Equipment type	O	1	e.g. **Home** or **MC**
13.4	Email Address	O		
13.5	Country Code	O		
13.6	Area/City Code	O		
13.7	Phone Number	O		
13.8	Extension	O		
13.9	Text	O		
14	Other Phone Number	O	1-n	
14.1	Number	O	1	
14.2	Telecoms Use code	O	1	e.g. **Internet**
14.3	Telecoms Equipment type	O	1	e.g. **WP**
14.4	Email Address	O		
14.5	Country Code	O		
14.6	Area/City Code	O		
14.7	Phone Number	O		
14.8	Extension	O		
14.9	Text	O		
15	Primary Language	O	1	
16	Marital Status	R	1	Code values accepted: **S** – Single **M** - Married/Civil Partner **D** - Divorced/Person whose Civil Partnership has been dissolved **W** - Widowed/Surviving Civil Partner **P** – Separated **N** - Not disclosed

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Seq	Field Name	Opt/Req	Occurs	Comments
				8 – Not applicable **9** – Not known Codes Values Accepted, but will be mapped to 9 – Not Known are: **O** **U** Codes Values Accepted, but will be mapped to P – Separated are: **E**
17	Religion	O	1	
19	SSN Number	O	1	
19.1	Social Security Number	O		
19.5	Number Type	O		NHS is recognised
22	Ethnic Group	O	1	The following look up values are used: **A** - British **B** - Irish **C** - Any other White background **D** - White and Black Caribbean **E** - White and Black African **F** - White and Asian **G** - Any other mixed background **H** - Indian **J** - Pakistani **K** - Bangladeshi **L** - Any other Asian background

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Seq	Field Name	Opt/Req	Occurs	Comments
				M - Caribbean **N** - African **P** - Any other Black background **R** - Chinese **S** - Any other ethnic group **Z** - Not stated If lookup value is not found, then original value that has been passed is retained.
28	Nationality	O	1	
29	Patient Death Date/Time	O	1	Should be blank unless the patient has died.
30	Patient Death Indicator	O	1	"Y" Indicates Yes
31	Patient Identity Unknown	O		
32	Identity Reliability Code	O	1	Populate this field if an NHS Number is supplied. CareCentric supports the following National Codes – **01** - Number present and verified **02** - Number present but not traced **03** - Trace required **04** - Trace attempted - No match or multiple match found **05** - Trace needs to be resolved **06** - Trace in progress **07** - Number not present and traced not required **08** – Trace Postponed
33	Last Update Date/Time	O		
34	Last Update Facility	O		
35	Species Code	O		
36	Breed Code	O		
37	Strain	O		

Seq	Field Name	Opt/Req	Occurs	Comments
38	Production Class Code	O		

5.4 PRD – Provider Data

Seq	Field Name	Opt/Req	Occurs	Comments
1.4	Specialty Code	O	1	
1.5	Specialty Description	O	1	
2.1	Consultant Surname	O	1	
2.2	Consultant Initials	O	1	
2.5	Consultant Title	O	1	
7.1	Referring Doctor Code	O	1	

5.5 PD1 – Patient Additional Demographic

Seq	Field Name	Opt/Req	Occurs	Comments
3	Patient Primary Facility	O	1	
3.1	Patient Primary Facility Name	O	1	GP Practice Name
3.3	Patient Primary Facility ID	O	1	GP Practice Code
4	Patient Primary Care Provider Name & ID No.	O	1	
4.1	GP ID Number	O	1	GP Code
4.2	GP Surname	O	1	
4.3	GP Forename	O	1	
4.6	GP Title	O	1	
4.7	GP ID Type	O	1	

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5.6 PV1 – Patient Visit Segment

Seq	Field Name	Opt/Req	Occurs	Comments
1	SetID	R	1	Set to '1'
2	PatientClass	R	1	Used to determine transaction type by: **AE** or **E** – Accident & Emergency **I** – Inpatient **O** – Outpatient **W** – Waiting List
3	Location	R	1	
3.1	Location Code	R	1	Either AE, Ward, Clinic or Waiting List Code
3.3	Location Bed	O	1	Specific Bed information
3.9	Location Description	R	1	Describes the location associated with PV1:3.1
4	Admission Type	O	1	
4.1	Admission Type Code	O	1	
4.2	Admission Type Description	O	1	
5	Preadmit Number	O		
7	Attending Doctor	R	1	
7.1	Code	R	1	i.e. GMC code
7.2	Surname	R	1	
7.3	Forename	O	1	
7.4	Initials	O	1	
7.6	Title	O	1	
8	Referring Doctor	R	1	
8.1	Code	R	1	i.e. GMC code
8.2	Surname	R	1	
8.3	Forename	O	1	

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Seq	Field Name	Opt/Req	Occurs	Comments
8.4	Initials	O	1	
8.6	Title	O	1	
9	Consulting Doctor / Physician	R	1-n	
9.1	Code	R	1	
9.2	Surname	R	1	
9.3	Forename	O	1	
9.4	Initials	O	1	
9.6	Title	O	1	
10	Specialty	O	1	
10.1	Specialty Code	O	1	
10.2	Specialty Description	O	1	
13	Re-Admission	O	1	
13.1	Re-Admission Code	O	1	
13.2	Re-Admission Description	O	1	
14	Admission Source	O	1	
14.1	Admission Code	O	1	
14.2	Admission Description	O	1	
17	Admitting Doctor	O	1	Required for Inpatient Admission messages
17.1	Code	R	1	
17.2	Surname	R	1	
17.3	Forename	O	1	
17.4	Initials	O	1	
17.6	Title	O	1	
18	Admission Category	O	1	
18.1	Admission Category Code	O	1	

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Seq	Field Name	Opt/Req	Occurs	Comments
18.2	Admission Category Description	O	1	
19	Patient Visit ID	R	1	Used to match activity
36	Discharge Method	O	1	Required for all Discharges
36.1	Discharge Method Code	O	1	
36.2	Discharge Method Description	O	1	
37	Discharge Destination	O	1	Required for all Discharges
37.1	Discharge Dest. Code	O	1	
37.2	Discharge Dest. Description	O	1	
44	Admission Date	O	1	Required for all Admissions
45	Discharge Date	O	1	Required for all Discharges

5.7 MRG – Merge

Seq	Field Name	Opt/Req	Occurs	Comments
1	Prior Patient Identifier List	R	1-n	
1.1	Patient ID	R	1	
1.4	Assigning Authority	O	1	Required for admission messages
1.5	Identity Type Code	R	1	Accepted Values **NHS** / **NH** – NHS Number **PAS** / **PI** - PAS or Hospital Number **HOSP** – Hospital Number **MRN** – Master Record Number
2	Prior Alternative Patient ID	O	0-n	
2.1	Patient ID	R	1	
2.4	Assigning Authority	O	1	

Seq	Field Name	Opt/Req	Occurs	Comments
2.5	Identity Type Code	R	1	Accepted Values **NHS** / **NH** – NHS Number **PAS** / **PI** - PAS or Hospital Number **HOSP** – Hospital Number **MRN** – Master Record Number
4	Prior Patient ID	O	0-n	
4.1	Patient ID	R	1	
4.4	Assigning Authority	O	1	
4.5	Identifying Type Code	R	1	Accepted Values **NHS** / **NH** – NHS Number **PAS** / **PI** - PAS or Hospital Number **HOSP** – Hospital Number **MRN** – Master Record Number
5	Prior Visit Number	O	0-n	
5.1	Patient ID	R	1	
5.3	Assigning Authority	O	1	
5.5	Identity Type Code	R	1	Accepted Values **NHS** / **NH** – NHS Number **PAS** / **PI** - PAS or Hospital Number **HOSP** – Hospital Number **MRN** – Master Record Number
6	Prior Alternative Visit ID	O	0-n	
6.1	Patient ID	R	1	
6.4	Assigning Authority	O	1	

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Seq	Field Name	Opt/Req	Occurs	Comments
6.5	Identity Type Code	R	1	Accepted Values **NHS** / **NH** – NHS Number **PAS** / **PI** - PAS or Hospital Number **HOSP** – Hospital Number **MRN** – Master Record Number
7	Prior Patient Name	O	1	

5.8 PV2 – Additional Patient Visit Segment

Seq	Field Name	Opt/Req	Occurs	Comments
3	Reason for Admission	O	1	
3.1	Reason for Admission Code	O	1	
3.2	Reason for Admission Desc.	O	1	Required for admission messages
4	Transfer Reason	O	1	Required for transfer messages
4.1	Transfer Reason Code	O	1	
4.2	Transfer Reason Description	O	1	
8	Admission Date	O	1	Required for all Pre-Admission Messages
9	Predicted Discharge Date	O	1	

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5.9 DG1 – Diagnosis Information

Seq	Field Name	Opt/Req	Occurs	Comments
1	SetID	R	1	To denote the iteration of DG1
3	Diagnosis	R	1	
3.1	Diagnosis Code	R	1	
3.2	Diagnosis Description	R	1	
3.3	Coding System	O	1	
5	Diagnosis Date	O	1	
6	Diagnosis Type	O	1	
15	Priority	O	1	
16	Clinician	O	1	
16.1	Code	R	1	
16.2	Surname	R	1	
16.3	Forename	O	1	
16.4	Initials	O	1	
16.6	Title	O	1	
16.8	Source	O	1	i.e. location
16.11	CheckDigit	O	1	

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5.10 PR1 – Procedure Information

Seq	Field Name	Opt/Req	Occurs	Comments
1	SetID	R	1	To denote the iteration of PR1
2	Procedure Coding Method	R	1	
3	Procedure Code	R	1	
3.1	Code	R	1	
3.2	Description	R	1	
3.3	Coding System	R	1	
4	Procedure Description	R	1	
5	Procedure Date Time	R	1	
6	Procedure Functional Type	O	1	
7	Procedure Minutes	O	1	length of time for procedure
8	Anaesthesiologist	O	1	
9	Anaesthesia Code	O	1	
10	Anaesthesia Minutes	O	1	
11	Surgeon	O	1	
12	Procedure Practitioner	O	1	
12.1	ID	O	1	
12.2	Surname	O	1	
12.3	Forename	O	1	
12.6	Title	O	1	
12.8	Source	O	1	
12.11	Check Digit	O	1	
13	Consent Code	O	1	
14	Procedure Priority	O	1	
15	Associated Diagnosis Code	O	1	

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Seq	Field Name	Opt/Req	Occurs	Comments
16	Procedure Code Modifier	O	1	
18	Tissue Type Code	O	1	

5.11 AL1 – Allergy Information

Seq	Field Name	Opt/Req	Occurs	Comments
1	Set ID	R	1	
2.1	Allergen Type Code / Alert Type code (see comments)	O	1	Allergies: **DA** - Drug Allergy **FA** - Food Allergy **MA** - Miscellaneous Allergy **MC** - Miscellaneous Contraindication **EA** - Environmental Allergy **AA** - Animal Allergy **PA** - Plant Allergy **LA** - Pollen Allergy Alerts: Any code or free text that does not match the above Allergy codes will be interpreted as an Alert Type code.
2.2	Alert Type Description	R	1	Description of Alert Type Only required when alert is sent.
3.1	Allergen Unique ID /Alert Sub-Type code (see comments)	R	1	Allergies: Uniquely Identifies the Allergy Alerts: Free text code for alerts sub-type.

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Seq	Field Name	Opt/Req	Occurs	Comments
3.2	Allergen Description/ Alert Sub-Type Description (see comments)	R	1	Allergies: Description of Allergy Alerts: Description of Alert Sub-Type
3.3	Allergen Coding System	O	1	
4.1	Severity	O	1	Code or Description **SV -** Severe **MO** - Moderate **MI -** Mild **U** - Unknown
5	Reaction / Alert Comments	O	1	Allergies: Code or description Alerts: Free text comments pertaining to the alert
6	Identification Date	R	1	Date allergy recorded

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5.12 RF1 – Referral Information

Seq	Field Name	Opt/Req	Occurs	Comments
1	Referral Outcome	O	1	
2	Referral Priority	O	1	
2.1	Code	R	1	
2.2	Description	R	1	
3	Referral Type	R	1	
3.1	Code	R	1	
3.2	Description	R	1	
4	Referral Disposition	O	1	
4.1	Code	R	1	
4.2	Description	R	1	
5	Referral Category	R	1	
5.1	Code	R	1	
5.2	Description	R	1	
6	Referral Identifier	R	1	
6.3	Identifier	R	1	
7	Effective Date\Time	R	1	
8	Expiration Date\Time	R	1	
9	Process Date\Time	R	1	
10	Referral Reason	O	1	
10.1	Code	R	1	
10.2	Description	R	1	
11	External Referral ID	O	1	
11.3	Identifier	R	1	

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5.13 PRD – Provider Data

Seq	Field Name	Opt/Req	Occurs	Comments
1	Admission Source	O	1	
1.1	Code	R	1	e.g. PP
1.2	Description	R	1	e.g. Primary Care Provider
2	Provider Name	R	1	
2.1	Surname	R	1	
2.2	Forename	O	1	
2.3	Initials	O	1	
2.5	Title	R	1	
3	Provider Address	R	1	
3.1	Street	R	1	
3.2	Other designation	O	1	
3.3	City / Town	O	1	
3.4	State / Province / County	O	1	
3.5	Post Code	R	1	
4	Provider Location	O	1	e.g. West Wing Red Clinic
5	Provider communication information	O	1	This field should contain information such as the phone number or email address used for communicating with the organisation
5.4	Email address	O	1	
5.7	Phone Number	O	1-n	
7	Provider Identifiers	R	1	Practice Code

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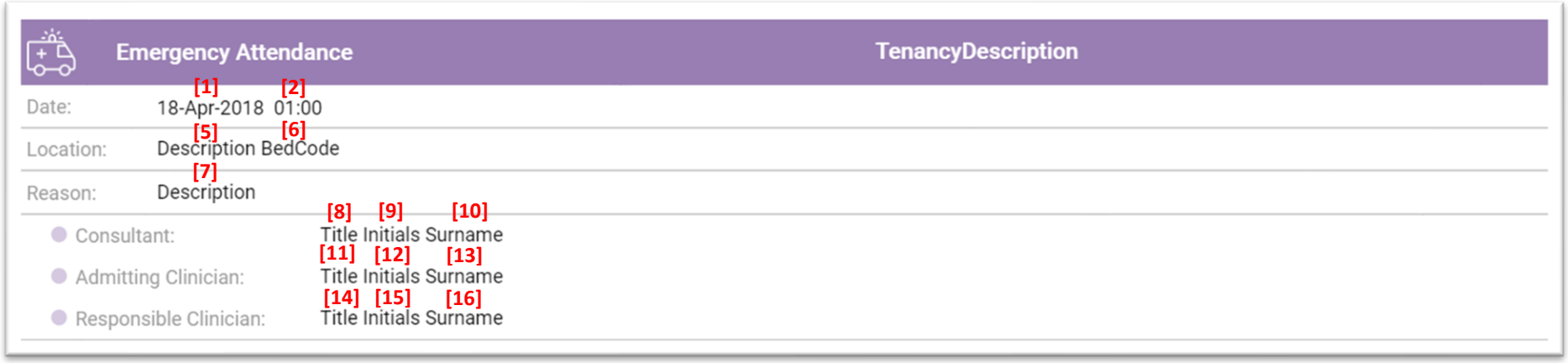
6 CareCentric ADT (including A&E)

When CareCentric Highway receives HL7 message types from the Trust Interface Engine / other 3rd party system, Highway v4 processes the messages as shown in CareCentric Highway Integration Engine. The data is then surfaced via Hub Tiles in the UI as shown in this section.

6.1 A&E Attendance (cc0024)

HL7 Message Type: ADT^A04 or ADT^A08
Patient Visit Class: AE or E or UniqueId starts with "EM"
Includes Demographics
Includes Allergies
Includes Alerts

6.1.1 Emergency Attendance Detail View V3



6.1.2 Emergency Attendance Detail View V4

Apply Sort/Filter

08-May-2024
Emergency Attendance
Prince William A&E
Prince William Foundation Trust

26-Apr-2024
Emergency Discharge
Wirral University A&E
Wirral University Teaching
Hospital NHS Foundation Trust

26-Apr-2024
Emergency Attendance
Wirral University A&E
Wirral University Teaching
Hospital NHS Foundation Trust

20-Apr-2024
Emergency Discharge
Prince William A&E
Prince William Foundation Trust

20-Apr-2024
Emergency Attendance
Prince William A&E
Prince William Foundation Trust

01-Mar-2024
Emergency Attendance
Orglinks A&E
Primary Care (GPs)

Emergency Activity - Prince William Foundation Trust

Other

Emergency Attendance

Realtime

[1] [2] 20-Apr-2024 10:29 — Attendance

[3] [4] 20-Apr-2024 14:29 — Discharge

Prince William A&E [5]

Accidental fall [7]

Discharge Method

PATIENT discharged on clinical advice or with clinical consent [17]

Discharge Destination

Usual place of residence unless listed below, for example, a private dwelling whether owner occupied or owned by Local Authority, housing association or other landlord. This includes wardened accommodation but not residential accommodation where health care is provided. It also includes PATIENTS with no fixed abode. [18]

People Involved

CLINICIAN	[8 9 10] Dr A Swanson
ADMITTING CLINICIAN	[11 12 13] Dr B Carole
RESPONSIBLE CLINICIAN	[14 15 16] Dr A Bernard

Coded Diagnosis

20-Apr-2024 10:29 [19]	Presenting Complaint ? Broken R leg [20]
20-Apr-2024 11:29	Closed fracture of tibia and fibula (disorder)

1 - 2 of 2

6.1.3 Mapping Reference

Reference	Description	HL7 Message Location	Term Name	Label
1	Attendance Date	PV1-44	ADT Date Time	Date
2	Attendance Time	PV1-44	ADT Date Time	Date
3	Discharge Date	PV1-45	ADT Date Time	Discharge
4	Discharge Time	PV1-45	ADT Date Time	Discharge
5	Location Description	PV1-3.9	ADT Emergency Attendance Discharge Location Description	Location
6	Bed (Code)	PV1-3.3	ADT Emergency Attendance Discharge Bed Code	Location
7	Reason for Attendance Description	PV2-3.2	ADT Admission Reason Description	Reason
8	Consultant Title	PV1-9.6	ADT Consultant Title	Consultant
9	Consultant Initials	PV1-9.3 & 9.4	ADT Consultant Initials	Consultant
10	Consultant Surname	PV1-9.2	ADT Consultant Surname	Consultant
11	Admitting Doctor – Title	PV1-17.6	ADT Admitting Doctor Title	Admitting Clinician
12	Admitting Doctor – Initials	PV1-17.3 & 17.4	ADT Admitting Doctor Initials	Admitting Clinician
13	Admitting Doctor – Surname	PV1-17.2	ADT Admitting Doctor Surname	Admitting Clinician
14	Attending Doctor – Title	PV1-7.6	ADT Attending Doctor Title	Responsible Clinician
15	Attending Doctor – Initials	PV1-7.3+7.4	ADT Attending Doctor Initials	Responsible Clinician
16	Attending Doctor – Surname	PV1-7.2	ADT Attending Doctor Surname	Responsible Clinician
17	Discharge Method Description	PV1-36.2	ADT Discharge Method Description	Method
18	Discharge Destination Description	PV1-37.2	ADT Discharge Destination Description	Destination
19	Diagnosis Date and Time	DG1.5	ADT Diagnosis Date Time	Under Coded

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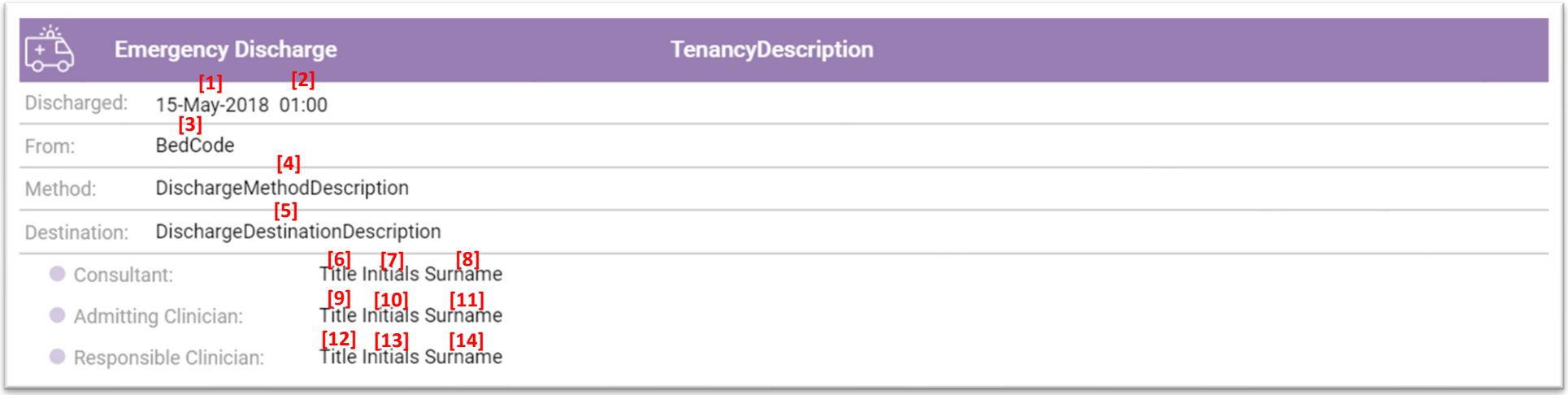
				Diagnosis
20	Diagnosis Description	DG1.4	ADT Diagnosis Description	Under Coded Diagnosis
21	Procedures Date and Time	PR1.5	ADT Procedures Date Time	Under Coded Procedures
22	Procedures Description	PR1.4	ADT Procedures Description	Under Coded Procedures

6.2 A&E Departure / Discharge (cc0025)

HL7 Message Type: ADT^A03
Patient Visit Class: AE or E or UniqueId starts with “EM”
Includes Demographics
Includes Allergies
Includes Alerts

6.2.1 Emergency Discharge Detail View V3



6.2.2 Emergency Discharge Detail View V4

Emergency Activity - Orglinks

Other

Emergency Discharge Realtime

[1] [2]
12-Dec-2015 16:10

Discharge

Windsor Ward [15]

Discharge Method

Patient discharged on clinical advice or with clinical consent [4]

Discharge Destination

Windsor Ward [5]

People Involved

CLINICIAN	[6 7 8] Dr S Wright
ADMITTING CLINICIAN	[9 10 11] Dr T Johnson
RESPONSIBLE CLINICIAN	[12 13 14] Dr T Johnson

Coded Diagnosis

13-Dec-2015 01:55 [16]	Fractured Left Hip [17]

1 - 1 of 1

6.2.3 Mapping Reference

Reference	Description	HL7 Message Location	Term Name	Label
1	Discharge Date	PV1-45	ADT Date Time	Discharged
2	Discharge Time	PV1-45	ADT Date Time	Discharged
3	Bed Code	PV1-3.3	ADT Discharge Code	From
4	Discharge Method Description	PV1-36.2	ADT Discharge Method Description	Method
5	Discharge Destination Description	PV1-37.2	ADT Discharge Destination Description	Destination
6	Consultant Title	PV1-9.6	ADT Consultant Title	Consultant
7	Consultant Initials	PV1-9.3 & 9.4	ADT Consultant Initials	Consultant
8	Consultant Surname	PV1-9.2	ADT Consultant Surname	Consultant
9	Admitting Doctor – Title	PV1-17.6	ADT Admitting Doctor Title	Admitting Clinician
10	Admitting Doctor – Initials	PV1-17.3 & 17.4	ADT Admitting Doctor Initials	Admitting Clinician
11	Admitting Doctor – Surname	PV1-17.2	ADT Admitting Doctor Surname	Admitting Clinician
12	Attending Doctor – Title	PV1-7.6	ADT Attending Doctor Title	Responsible Clinician
13	Attending Doctor – Initials	PV1-7.3 & 7.4	ADT Attending Doctor Initials	Responsible Clinician
14	Attending Doctor – Surname	PV1-7.2	ADT Attending Doctor Surname	Responsible Clinician
15	Location Description	PV1.3.9	ADT Location Description	-
16	Diagnosis Date and Time	DG1.5	ADT Diagnosis Date Time	Under Coded Diagnosis
17	Diagnosis Description	DG1.4	ADT Diagnosis Description	Under Coded Diagnosis
	Procedures Date and Time	PR1.5	ADT Procedures Date Time	Under Coded Procedures
	Procedures Description	PR1.4	ADT Procedures Description	Under Coded Procedures

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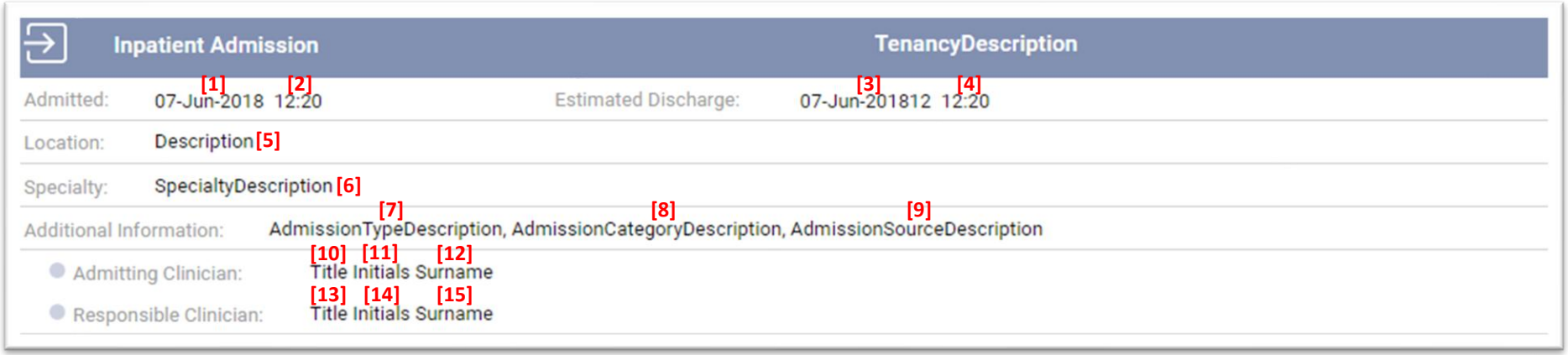
Inpatient Admission (cc0014)
HL7 Message Type:
ADT^A01 & Event 100
ADT^A08 & Event 101
ADT^A08 & Event 121 or 131 with UniqueID ending with "-"
Patient Visit Class: I or UniqueID starts with "IP"
Includes Demographics
Includes Allergies
Includes Alerts

6.2.4 Inpatient Admission Detail View V3



6.2.5 Inpatient Admission Detail View V4

Inpatient Activity - Prince William Foundation Trust

Inpatient Admission Realtime

[1] [2] [3] [4]

15-Jan-2024 13:01 › Not provided

Admission — Expected Discharge

Location

Ward	DAY SURGICAL UNIT	[5]
Bed	3c	[19]

Specialty

GENERAL SURGERY [6]

Admission Information

Category	NHS PATIENT, including Overseas Visitors charged under the National Health Service (Overseas Visitors Hospital Charging Regulations)	[8]
Type	Elective Admission: Waiting list	[7]
Source	Usual place of residence	[9]

People Involved

CLINICIAN	[16 17 18] Mr James Stokes
ADMITTING CLINICIAN	[10 11 12] Mr Reg Istrar
RESPONSIBLE CLINICIAN	[13 14 15] Mr C Onsultant

6.2.6 Mapping Reference

Reference	Description	HL7 Message Location	Term Name	Label
1	Admission Date	PV1-44	ADT Date Time	Admitted
2	Admission Time	PV1-44	ADT Date Time	Admitted
3, 4	Expected Discharge Date	PV2-9	ADT Expected Discharge Date	Estimated Discharge
5	Ward Description	PV1-3.9	ADT Ward Description	Location
6	Specialty Description	PV1-10.2	ADT Specialty Description	Specialty
7	Admission Type Description	PV1-4.2	ADT Admission Type Description	Additional Information
9	Admission Source Description	PV1-14.2	ADT Admission Source Description	Additional Information
8	Admission Category Description	PV1-18.2	ADT Admission Category Description	Additional Information
10	Admitting Doctor – Title	PV1-17.6	ADT Admitting Doctor Title	Admitting Clinician
11	Admitting Doctor – Initials	PV1-17.3 & 17.4	ADT Admitting Doctor Initials	Admitting Clinician
12	Admitting Doctor – Surname	PV1-17.2	ADT Admitting Doctor Surname	Admitting Clinician
13	Attending Doctor – Title	PV1-7.6	ADT Attending Doctor Title	Responsible Clinician
14	Attending Doctor – Initials	PV1-7.3 & 7.4	ADT Attending Doctor Initials	Responsible Clinician
15	Attending Doctor – Surname	PV1-7.2	ADT Attending Doctor Surname	Responsible Clinician
16	Consultant – Title	PV1-9.6	ADT Consultant Title	Clinician
17	Consultant – Initials	PV1-9.3 & 9.4	ADT Consultant Initials	Clinician
18	Consultant - Surname	PV1-9.2	ADT Consultant Surname	Clinician
19	Bed Code	PV1-3.3	ADT Bed Code	Bed

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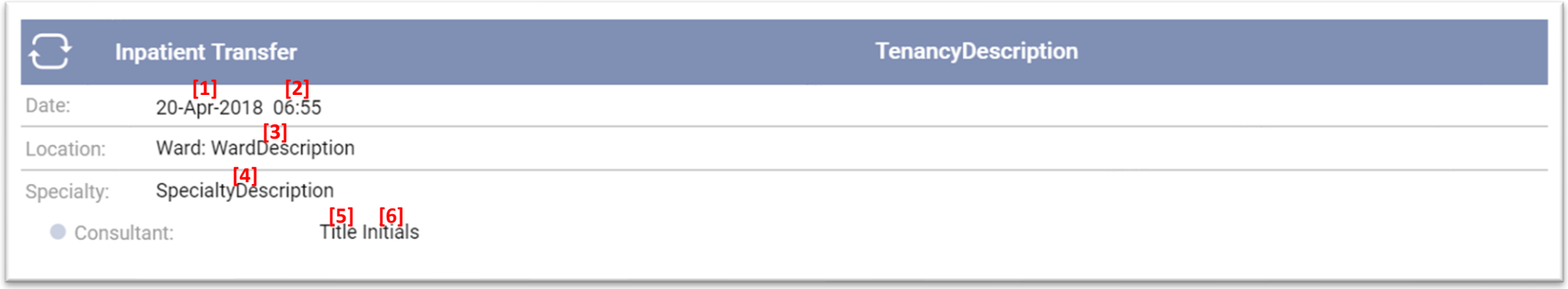
6.3 Inpatient Transfer (cc0016)

HL7 Message Type:
ADT^A02 & Event 120 or 130
ADT^A08 & Event 121 or 131 and UniqueID contains "-" but doesn't start with or end with "-"
Patient Visit Class: I or UniqueID starts with "IP"

6.3.1 Inpatient Transfer Detail View V3



6.3.2 Inpatient Transfer (Ward Transfer and Consultant Update) Detail View V4

Apply Sort/Filter

24-Sep-2022
Inpatient Transfer
GENERAL INTERNAL MEDICINE
Prince William Foundation Trust

23-Sep-2022
Inpatient Admission
GENERAL INTERNAL MEDICINE
Prince William Foundation Trust

27-Sep-2021
Inpatient Discharge
GENERAL INTERNAL MEDICINE
Prince William Foundation Trust

Inpatient Activity - Prince William Foundation Trust
Inpatient Transfer (Ward Transfer) Realtime

[1] [2]
24-Sep-2022 17:19
Transfer

Reason
No data available [9]

Location
Ward Ward 12 [3]
Bed Bed 12 [8]

Inpatient Activity - Prince William Foundation Trust
Inpatient Transfer Consultant Update Realtime Other

[1] [2]
01-May-2024 12:30
Transfer

Reason
Reason 1 [9]

Specialty
General Surgery [4]

People Involved
[5][6] [7]
CLINICIAN Mr B Greens

6.3.3 Mapping Reference

Reference	Description	HL7 Message Location	Term Name	Label
1	Transfer Date	EVN-6	ADT Date Time	Date
2	Transfer Time	EVN-6	ADT Date Time	Date
3	Ward Description	PV1-3.9	ADT Transfer Ward Description	Location
4	Specialty Description	PV1-10.2	ADT Transfer Specialty Description	Specialty
5	Consultant Title	PV1-9.6	ADT Transfer Consultant Title	Consultant/ Clinician
6	Consultant Initials	PV1-9.3 & 9.4	ADT Transfer Consultant Initials	Consultant/ Clinician
7	Consultant Surname	PV1-9.2	ADT Transfer Consultant Surname	Consultant/ Clinician
8	Bed Code	PV1-3.3	ADT Bed Code	Bed
9	Transfer Reason	PV2-4	ADT Transfer Reason	Reason

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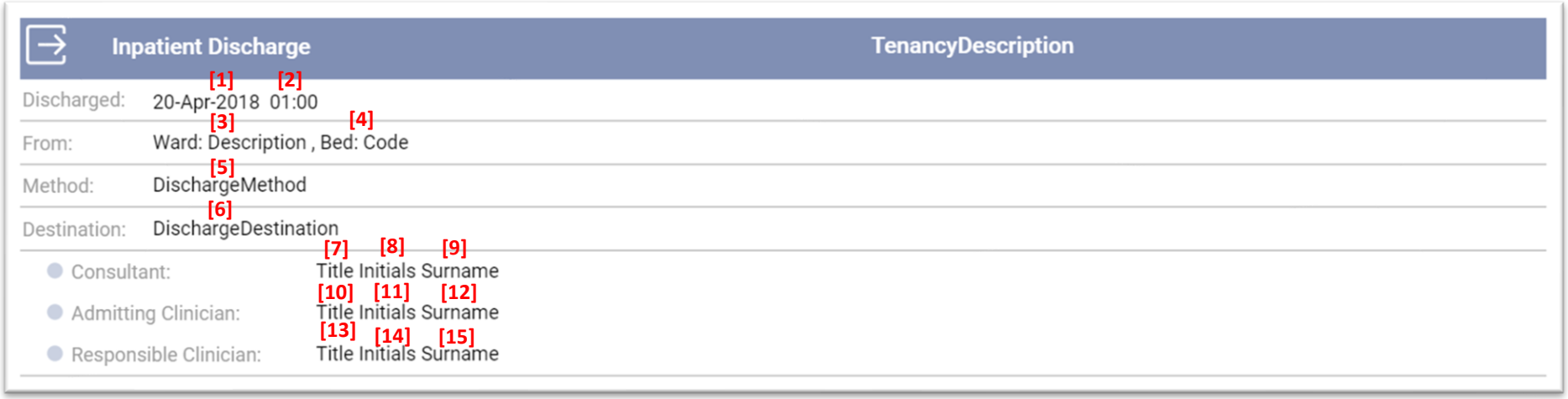
6.4 Inpatient Discharge (cc0015)

HL7 Message Type:
ADT^A03 & Event 140
ADT^A08 & Event 141
Patient Visit Class: I or UniqueID starts with “IP”
Includes Demographics
Includes Allergies
Includes Alerts

6.4.1 Inpatient Discharge Detail View V3



6.4.2 Inpatient Discharge Detail View V4

Apply Sort/Filter

02-May-2024
Inpatient Discharge
General Surgery
Prince William Foundation Trust

01-May-2024
Inpatient Transfer
General Surgery
Prince William Foundation Trust

01-May-2024
Inpatient Transfer
General Surgery
Prince William Foundation Trust

01-May-2024
Inpatient Transfer
General Surgery
Prince William Foundation Trust

01-May-2024
Inpatient Transfer
General Surgery
Prince William Foundation Trust

Inpatient Activity - Prince William Foundation Trust

Inpatient Discharge

Realtime

[1] [2]
02-May-2024 11:29

Discharge

Location

Ward	Rutherford Ward [3]
Bed	04 [4]

Discharge Information

Method	Discharged - follow up treatment to be provided by GENERAL PRACTITIONER [5]
Destination	Usual place of residence unless listed below, for example, a private dwelling whether owner occupied or owned by Local Authority, housing association or other landlord. This includes wardened accommodation but not residential accommodation where health care is provided. It also includes PATIENTS with no fixed abode [6]

People Involved

CLINICIAN	[7 8 9] MR B Green
ADMITTING CLINICIAN	[10 11 12] Dr E Bacon
RESPONSIBLE CLINICIAN	[13 14 15] Dr S Smith

6.4.3 Mapping Reference

Reference	Description	HL7 Message Location	Term Name	Label
1	Discharge Date	PV1-45	ADT Date Time	Discharged
2	Discharge Time	PV1-45	ADT Date Time	Discharged
3	Ward Description	PV1-3.9	ADT Discharge Ward Description	From
4	Bed	PV1-3.3	ADT Discharge Bed Code	From
5	Discharge Method Description	PV1-36.2	ADT Discharge Method Description	
6	Discharge Destination Description	PV1-37.2	ADT Discharge Destination Description	
7	Consultant Title	PV1-9.6	ADT Consultant Title	Consultant/ Clinician
8	Consultant Initials	PV1-9.3 & 9.4	ADT Consultant Initials	Consultant/ Clinician
9	Consultant Surname	PV1-9.2	ADT Consultant Surname	Consultant/ Clinician
10	Admitting Doctor – Title	PV1-17.6	ADT Admitting Doctor Title	Admitting Clinician
11	Admitting Doctor – Initials	PV1-17.3 & 17.4	ADT Admitting Doctor Initials	Admitting Clinician
12	Admitting Doctor – Surname	PV1-17.2	ADT Admitting Doctor Surname	Admitting Clinician
13	Attending Doctor – Title	PV1-7.6	ADT Attending Doctor Title	Responsible Clinician
14	Attending Doctor – Initials	PV1-7.3 & 7.4	ADT Attending Doctor Initials	Responsible Clinician
15	Attending Doctor – Surname	PV1-7.2	ADT Attending Doctor Surname	Responsible Clinician

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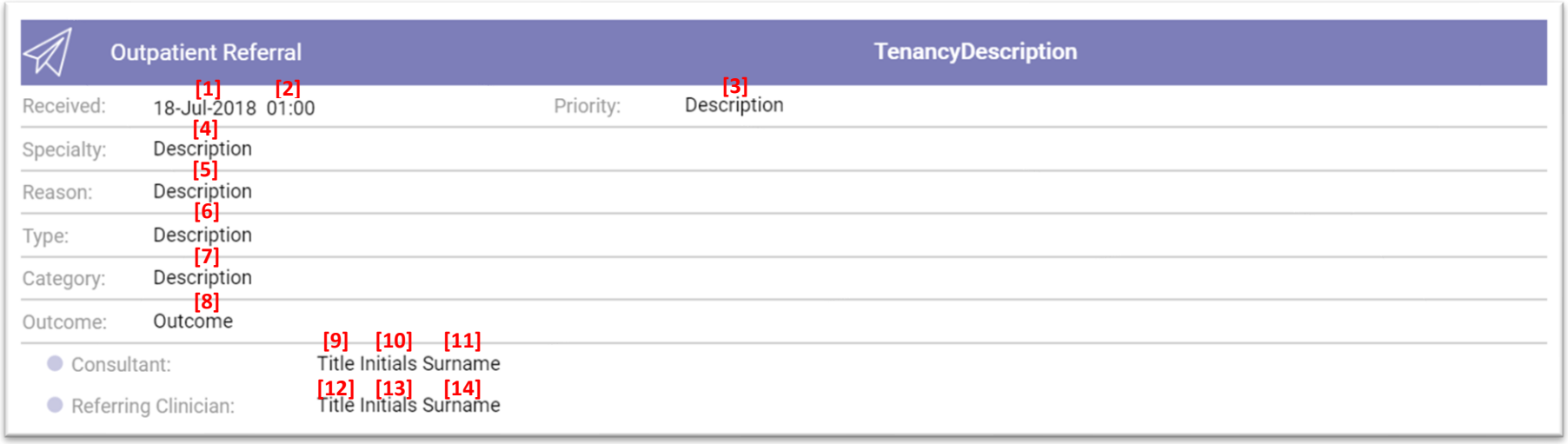
6.5 Outpatient Referral (cc0019)

HL7 Message Type: REF^I12
Patient Visit Class: Anything other than W
Includes Demographics
Includes Allergies
Includes Alerts

6.5.1 Outpatient Referral Detail View V3



6.5.2 Outpatient Referral Detail View V4

Apply Sort/Filter

02-Jun-2025
Outpatient Appointment
OPHTHALMOLOGY
Prince William NHS Foundation Trust

04-May-2025
Outpatient Referral
OPHTHALMOLOGY
Prince William NHS Foundation Trust

09-Jan-2024
Outpatient Attendance
GENERAL SURGERY
Prince William NHS Foundation Trust

04-Jan-2024
Outpatient Attendance
GENERAL SURGERY
Prince William NHS Foundation Trust

Outpatient Activity - Prince William NHS Foundation Trust

Outpatient Referral Realtime

[1] [2]
04-May-2025 09:00

Received

Priority

No data available [3]

Specialty

OPHTHALMOLOGY [4]

Referral Information

Reason	No data available [5]
Type	No data available [6]
Category	No data available [7]
Outcome	No data available [8]

People Involved

CLINICIAN	[9 10 11] Mr David Percival
REFERRED BY	[12 13 14] Dr Am Reyes-Luna

6.5.3 Mapping Reference

Reference	Description	HL7 Message Location	Term Name	Label
1, 2	Referral Date	RF1-7	ADT Date Time	Received
3	Referral Priority Description	RF1-2.2	ADT Referral Priority Description	Priority
4	Specialty Description	PV1-10.2	ADT Specialty Description	Specialty
5	Referral Reason Description	RF1-10.2	ADT Referral Reason Description	Reason
6	Referral Type Description	RF1-3.2	ADT Referral Type Description	Type
7	Referral Category Description	RF1-5.2	ADT Referral Category Description	Category
8	Referral Outcome	RF1-1.1	ADT Admission Referral Outcome	Outcome
9	Consultant Title	PV1-9.6	ADT Consultant Title	Consultant/Clinician
10	Consultant Initials	PV1-9.3 & 9.4	ADT Consultant Initials	Consultant/Clinician
11	Consultant Surname	PV1-9.2	ADT Consultant Surname	Consultant/Clinician
12	Referring Doctor – Title	PV1-8.6	ADT Referring Doctor Title	Referring Clinician
13	Referring Doctor – Initials	PV1-8.3 & 8.4	ADT Referring Doctor Initials	Referring Clinician
14	Referring Doctor – Surname	PV1-8.2	ADT Referring Doctor Surname	Referring Clinician

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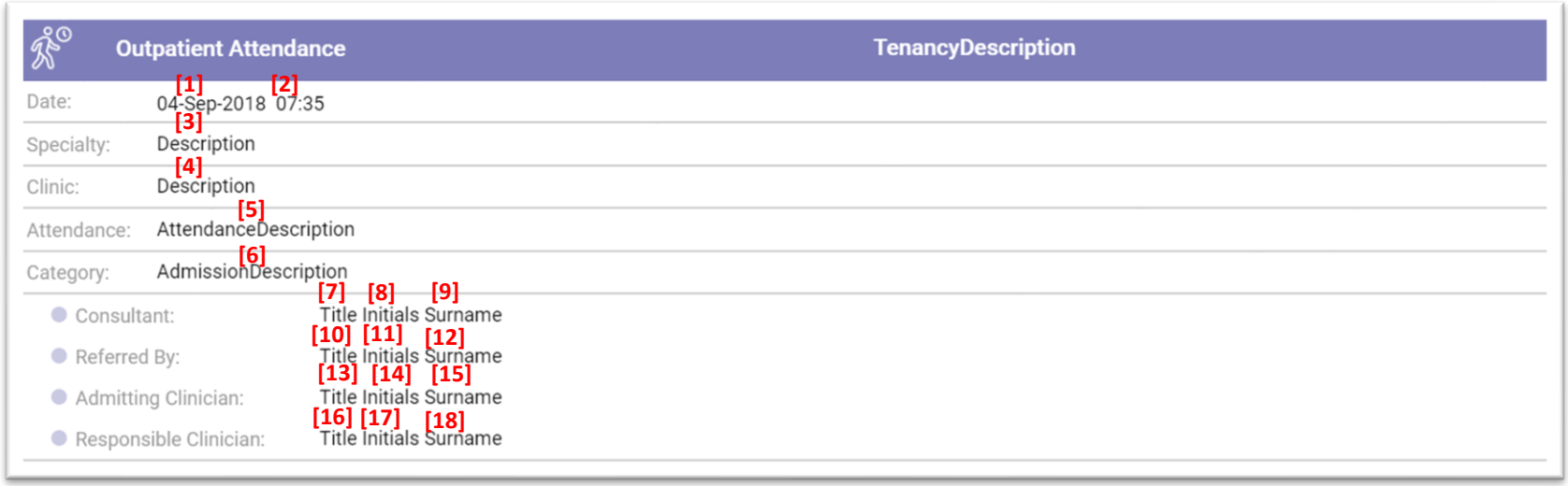
6.6 Outpatient Attendance (cc0027)

HL7 Message Type:
ADT^A01
ADT^A04
ADT^A08
Patient Visit Class: O or UniqueID starts with “OP”
Includes Demographics
Includes Allergies
Includes Alerts

6.6.1 Outpatient Attendance Detail View V3



6.6.2 Outpatient Attendance Detail View V4

Outpatient Activity - Prince William Foundation Trust

Outpatient Attendance Realtime

[1] [2]
22-Feb-2023 11:29

Visit Date

Specialty
Medical Oncology [3]

Clinic
Dr B Plum - OPWL [4]

Attendance

Type	Appointment first attendance [5]
Category	NHS PATIENT, including Overseas Visitors charged under the National Health Service (Overseas Visitors Hospital Charging Regulations) [6]

People Involved

CLINICIAN	[7 8 9] Dr B Plum
REFERRED BY	[10 11 12] Dr B Smith
ADMITTING CLINICIAN	[13 14 15] Dr D Schmit
RESPONSIBLE CLINICIAN	[16 17 18] Dr F Spend

6.6.3 Mapping Reference

Reference	Description	HL7 Message Location	Term Name	Label
1	Visit Date	PV1-44	ADT Date Time	Date
2	Visit Time	PV1-44	ADT Date Time	Date
3	Specialty Description	PV1-10.2	ADT Specialty Description	Specialty
4	Location/Clinic	PV1-3.9 & 3.1	ADT Location Description + ADT Location Code	Clinic
5	Attendance Type Description	PV1-4.2	ADT Outpatient Attendance Type Description	Attendance
6	Attendance Category Description	PV1-18.2	ADT Admission Category Description	Category
7	Consultant Title	PV1-9.6	ADT Consultant Title	Consultant/ Clinician
8	Consultant Initials	PV1-9.3 & 9.4	ADT Consultant Initials	Consultant/ Clinician
9	Consultant Surname	PV1-9.2	ADT Consultant Surname	Consultant/ Clinician
10	Referring Doctor – Title	PV1-8.6	ADT Referring Doctor Title	Referred By
11	Referring Doctor – Initials	PV1-8.3 & 8.4	ADT Referring Doctor Initials	Referred By
12	Referring Doctor – Surname	PV1-8.2	ADT Referring Doctor Surname	Referred By
13	Admitting Doctor – Title	PV1-17.6	ADT Admitting Doctor Title	Admitting Clinician
14	Admitting Doctor – Initials	PV1-17.3 & 17.4	ADT Admitting Doctor Initials	Admitting Clinician
15	Admitting Doctor – Surname	PV1-17.2	ADT Admitting Doctor Surname	Admitting Clinician
16	Attending Doctor – Title	PV1-7.6	ADT Attending Doctor Title	Responsible Clinician
17	Attending Doctor – Initials	PV1-7.3 & 7.4	ADT Attending Doctor Initials	Responsible Clinician
18	Attending Doctor – Surname	PV1-7.2	ADT Attending Doctor Surname	Responsible Clinician

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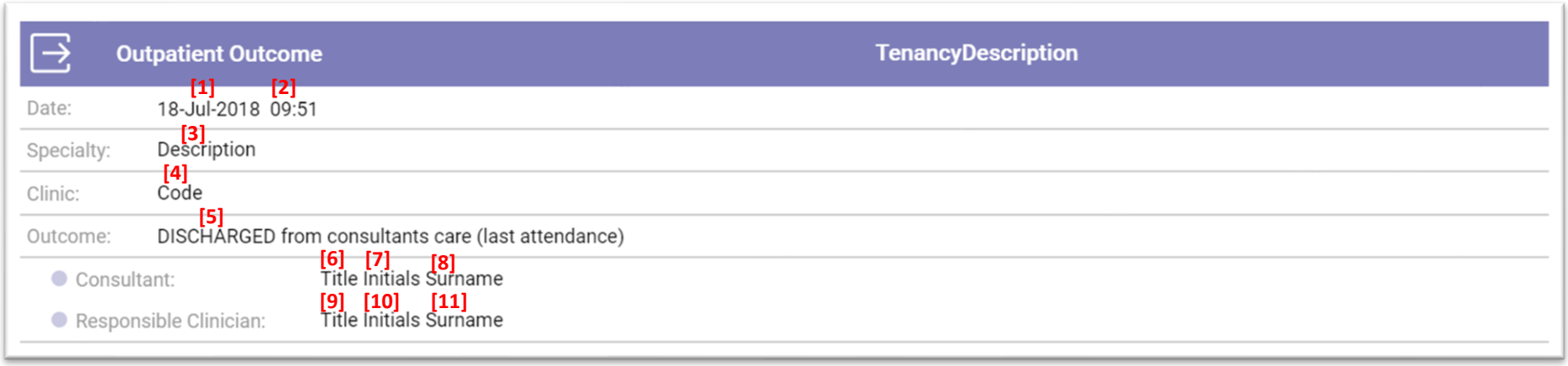
6.7 Outpatient Outcome (cc0021)

HL7 Message Type: ADT^A03
Patient Visit Class: O or UniqueID starts with “OP”
Includes Demographics
Includes Allergies
Includes Alerts

6.7.1 Outpatient Outcome Detail View V3



6.7.2 Outpatient Outcome Detail View V4

Outpatient Activity - Prince William Foundation Trust

Outpatient Outcome Realtime

[1] [2]
20-Aug-2024 11:29

Discharge Date

Outcome

- FOLLOW UP appointment given [5]

Specialty

General Surgery [3]

Clinic

Dr C James - OPWL [4]

People Involved

CLINICIAN	[6 7 8] Dr C James
RESPONSIBLE CLINICIAN	[9 10 11] Dr T Trent

6.7.3 Mapping Reference

Reference	Description	HL7 Message Location	Term Name	Label
1	Discharge Date	PV1-45	ADT Date Time	Date
2	Discharge Time	PV1-45	ADT Date Time	Date
3	Specialty Description	PV1-10.2	ADT Specialty Description	Specialty
4	Location/Clinic	PV1-3.9 & 3.1	ADT Location Description + ADT Location Code	Clinic
5	Discharge Method Code	PV1-36.1	ADT Discharge Method Code	Outcome
6	Consultant Title	PV1-9.6	ADT Consultant Title	Consultant
7	Consultant Initials	PV1-9.3 & 9.4	ADT Consultant Initials	Consultant
8	Consultant Surname	PV1-9.2	ADT Consultant Surname	Consultant
9	Attending Doctor – Title	PV1-7.6	ADT Attending Doctor Title	Attending Clinician
10	Attending Doctor – Initials	PV1-7.3 & 7.4	ADT Attending Doctor Initials	Attending Clinician
11	Attending Doctor – Surname	PV1-7.2	ADT Attending Doctor Surname	Attending Clinician

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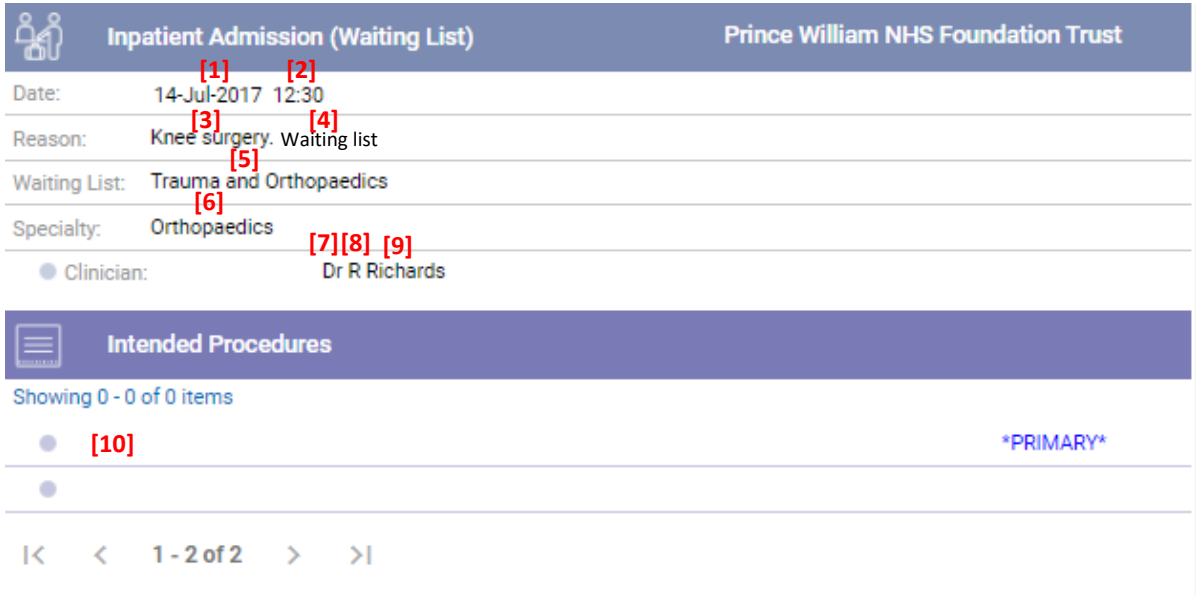
6.8 Inpatient Admission (Waiting List) (cc0017)

HL7 Message Type: ADT^A05 or ADT^A08
Patient Visit Class: I or W
Includes Demographics
Includes Allergies
Includes Alerts

6.8.1 Inpatient Admission (Waiting List) V3



6.8.2 Inpatient Admission (Waiting List) V4

Inpatient Activity - Prince William Foundation Trust

Inpatient Waiting List Realtime

[1] [2]
30-Apr-2023 11:29
Date

Specialty
General Surgery [6]

Waiting List Information

Reason	Elective Admission: Waiting list [3]
Type	Waiting list [4]
Waiting List	Mr R Jones - IPWL [5]

People Involved

[7 8 9]

CLINICIAN	Mr R Jones

Procedures

PRIMARY	Exploratory laparotomy [10]
	Surgical Biopsy

1 - 2 of 2

6.8.3 Mapping Reference

Reference	Description	HL7 Message Location	Term Name	Label
1,2	Admission Date Time	PV2-8	ADT Admission Date	Date
3	Reason for Admission Description	PV2-3.2	ADT Admission Reason Description	Reason
4	Admission Type Description	PV1-4.2	ADT Admission Type Description	Reason
5	Ward Description	PV1-3.9	ADT Ward Description	Waiting List
6	Specialty Description	PV1-10.2	ADT Specialty Description	Specialty
7	Consultant Title	PV1-9.6	ADT Consultant Title	Clinician
8	Consultant Initials	PV1-9.3 & 9.4	ADT Consultant Initials	Clinician
9	Consultant Surname	PV1-9.2	ADT Consultant Surname	Clinician
10	Procedure Description	PR1-3.2	ADT Procedure Code	-

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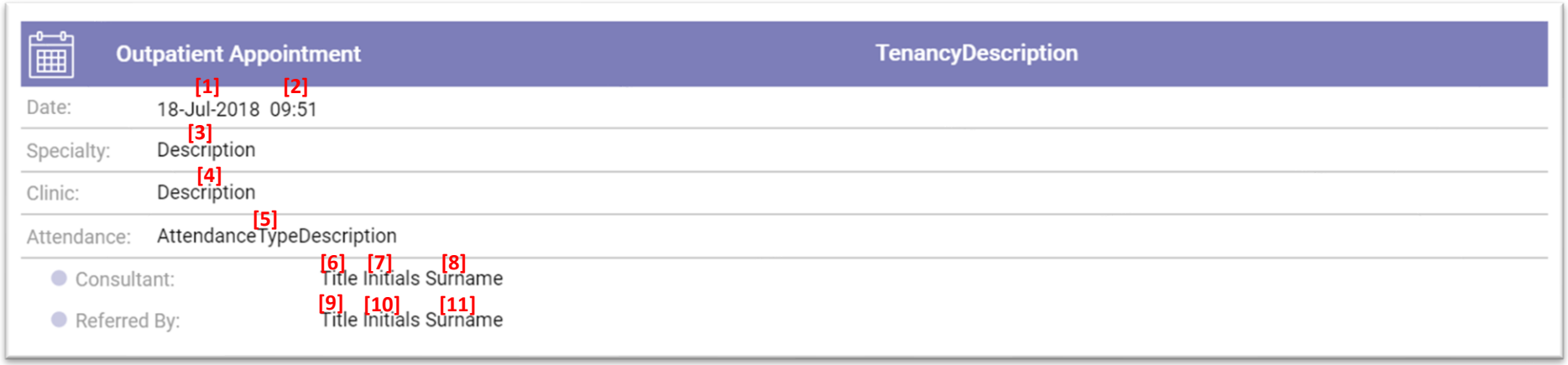
6.9 Outpatient Pre-Appointment (cc0020)

HL7 Message Type: ADT^A05
Patient Visit Class: O or UniqueID starts with “OP”

6.9.1 Outpatient Pre-Appointment Detail View V3



6.9.2 Outpatient Pre-Appointment Detail View V4

Outpatient Activity - Prince William Foundation Trust

Outpatient Appointment Realtime

[1] [2]
20-Jan-2025 11:29

Planned Attendance Date

Specialty
General Surgery [3]

Clinic
Dr R James - OPWL [4]

Attendance Type
Appointment first attendance [5]

People Involved

CLINICIAN	[6 7 8] Dr R James
REFERRED BY	[9 10 11] Dr A Jones

6.9.3 Mapping Reference

Reference	Description	HL7 Message Location	Term Name	Label
1	Planned Attendance Date	PV2-8	ADT Admission Date	Date
2	Planned Attendance Time	PV1-44	ADT Date Time	Date
3	Specialty Description	PV1-10.2	ADT Specialty Description	Specialty
4	Location/Clinic	PV1-3.9 & 3.1	ADT Outpatient Clinic Code + ADT Outpatient Clinic Description	Clinic
5	Attendance Type Description	PV1-4.2	ADT Outpatient Attendance Type Description	Attendance
6	Consultant Title	PV1-9.6	ADT Consultant Title	Consultant
7	Consultant Initials	PV1-9.3 & 9.4	ADT Consultant Initials	Consultant
8	Consultant Surname	PV1-9.2	ADT Consultant Surname	Consultant
9	Referring Doctor – Title	PV1-8.6	ADT Referring Doctor Title	Referred By
10	Referring Doctor – Initials	PV1-8.3 & 8.4	ADT Referring Doctor Initials	Referred By
11	Referring Doctor – Surname	PV1-8.2	ADT Referring Doctor Surname	Referred By

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7 Mapping Considerations

7.1 Document Properties

Document properties (or Meta Data relating to the document) are stored separately from the document XML in CareCentric. Each HL7 message type will be expected to contain the following data and stored in the database:

Database Table – Column	HL7 Message Location
[dbo].[DocumentProperties] – Message From	//MSH/MSH.3/MSH.3.1
[dbo].[DocumentProperties] – Message ID	//MSH/MSH.10/MSH.10.1
[dbo].[DocumentProperties] – Message Type	//MSH/MSH.9/MSH.9.1
[dbo].[DocumentProperties] – Published Date	This is the date on which the message is processed by Highway.

7.2 CareCentric Version Control

Version control is required for all messages

7.2.1 ADT Messages

For A&E, IP, OP and WL messages documents will be version controlled based on the **Patient Visit ID** which is retrieved from the HL7 location: **//PV1/PV1.19.1.**

For MPI messages (i.e. A28, A31 and A40), messages will only apply to patient demographics and will therefore not link back to a specific document within Graphnet.

7.3 CareCentric Business Rules

The following business rules will be used when processing the data for creating, updating and deleting data.

ADT Documents	
Create	New records based on the key **Patient Visit ID** will be detected and added. The identifier will be retained in the *UniqueID* column of the Graphnet document table
Update	Existing records based on the key **Patient Visit ID** will be detected and the inbound document will be made a newer version of the existing document. HL7 message types ADT A08 and REF I13 will indicate an update. Other HL7 message types with the same message control ID will also be treated as an update.
Delete	Data is stored and updated, never deleted. Cancellations will hide data from the patient record (soft delete).

8 Patient Merges

When sending data for merged patient records, the update should contain the Patient ID for the record the change has been made against, whether that is the master or minor number. Any updates against the master or minor record will appear as part of the master record in CareCentric but will retain an association to the original record should there be any requirement to demerge in future.